

# General Purpose Transistors

## NPN Silicon

### FEATURE

- High current capacity in compact package.  
 $I_C = 0.8A$ .
- Epitaxial planar type.
- PNP complement: L8550
- Pb-Free Package is available.

### DEVICE MARKING AND ORDERING INFORMATION

| Device     | Marking          | Shipping       |
|------------|------------------|----------------|
| L8050PLT1  | 80P              | 3000/Tape&Reel |
| L8050PLT1G | 80P<br>(Pb-Free) | 3000/Tape&Reel |
| L8050QLT1  | 1YC              | 3000/Tape&Reel |
| L8050QLT1G | 1YC<br>(Pb-Free) | 3000/Tape&Reel |

### MAXIMUM RATINGS

| Rating                       | Symbol    | Max | Unit |
|------------------------------|-----------|-----|------|
| Collector-Emitter Voltage    | $V_{CEO}$ | 25  | V    |
| Collector-Base Voltage       | $V_{CBO}$ | 40  | V    |
| Emitter-Base Voltage         | $V_{EBO}$ | 5   | V    |
| Collector Current-continuoun | $I_C$     | 800 | mAdc |

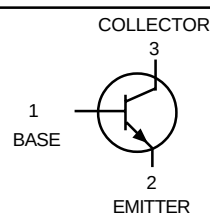
### THERMAL CHARACTERISTICS

| Characteristic                                                                                    | Symbol          | Max         | Unit                  |
|---------------------------------------------------------------------------------------------------|-----------------|-------------|-----------------------|
| Total Device Dissipation FR-5 Board,(1)<br>$T_A=25^{\circ}C$<br>Derate above $25^{\circ}C$        | $P_D$           | 225<br>1.8  | mW<br>mW/ $^{\circ}C$ |
| Thermal Resistance,Junction to Ambient                                                            | $R_{\theta JA}$ | 556         | $^{\circ}C/W$         |
| Total Device Dissipation<br>Alumina Substrate,(2) $T_A=25^{\circ}C$<br>Derate above $25^{\circ}C$ | $P_D$           | 300<br>2.4  | mW<br>mW/ $^{\circ}C$ |
| Thermal Resistance,Junction to Ambient                                                            | $R_{\theta JA}$ | 417         | $^{\circ}C/W$         |
| Junction and Storage Temperature                                                                  | $T_j, T_{stg}$  | -55 to +150 | $^{\circ}C$           |

### L8050\*LT1



SOT-23



**L8050\*LT1**
**ELECTRICAL CHARACTERISTICS ( $T_A=25^{\circ}\text{C}$  unless otherwise noted)**

| Characteristic | Symbol | Min | Typ | Max | Unit |
|----------------|--------|-----|-----|-----|------|
|----------------|--------|-----|-----|-----|------|

**OFF CHARACTERISTICS**

|                                                               |               |    |   |     |    |
|---------------------------------------------------------------|---------------|----|---|-----|----|
| Collector-Emitter Breakdown Voltage<br>( $I_C=1.0\text{mA}$ ) | $V_{(BR)CEO}$ | 25 | — | —   | V  |
| Emitter-Base Breakdown Voltage<br>( $I_E=100\mu\text{A}$ )    | $V_{(BR)EBO}$ | 5  | — | —   | V  |
| Collector-Base Breakdown Voltage<br>( $I_C=100\mu\text{A}$ )  | $V_{(BR)CBO}$ | 40 | — | —   | V  |
| Collector Cutoff Current ( $V_{CB}=35\text{V}$ )              | $I_{CBO}$     | —  | — | 150 | nA |
| Emitter Cutoff Current ( $V_{EB}=4\text{V}$ )                 | $I_{EBO}$     | —  | — | 150 | nA |

**ELECTRICAL CHARACTERISTICS ( $T_A=25^{\circ}\text{C}$  unless otherwise noted)**

| Characteristic | Symbol | Min | Typ | Max | Unit |
|----------------|--------|-----|-----|-----|------|
|----------------|--------|-----|-----|-----|------|

**ON CHARACTERISTICS**

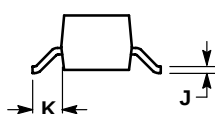
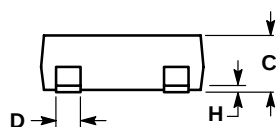
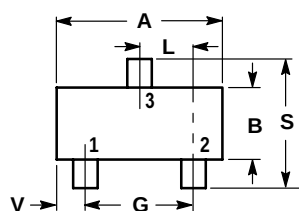
|                                                                |             |     |   |     |   |
|----------------------------------------------------------------|-------------|-----|---|-----|---|
| DC Current Gain<br>$I_C=100\text{mA}, V_{CE}=1\text{V}$        | $h_{FE}$    | 150 | - | 600 |   |
| Collector-Emitter Saturation Voltage<br>( $I_C=800\text{mA}$ ) | $V_{CE(S)}$ | -   | - | 0.5 | V |

**NOTE :**

| *        | P       | Q       | R       | S       |
|----------|---------|---------|---------|---------|
| $h_{FE}$ | 100~200 | 150~300 | 200~400 | 300~600 |

**L8050\*LT1**
**SOT-23**
**NOTES:**

1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M,1982
2. CONTROLLING DIMENSION: INCH.



| DIM | INCHES |        | MILLIMETERS |       |
|-----|--------|--------|-------------|-------|
|     | MIN    | MAX    | MIN         | MAX   |
| A   | 0.1102 | 0.1197 | 2.80        | 3.04  |
| B   | 0.0472 | 0.0551 | 1.20        | 1.40  |
| C   | 0.0350 | 0.0440 | 0.89        | 1.11  |
| D   | 0.0150 | 0.0200 | 0.37        | 0.50  |
| G   | 0.0701 | 0.0807 | 1.78        | 2.04  |
| H   | 0.0005 | 0.0040 | 0.013       | 0.100 |
| J   | 0.0034 | 0.0070 | 0.085       | 0.177 |
| K   | 0.0140 | 0.0285 | 0.35        | 0.69  |
| L   | 0.0350 | 0.0401 | 0.89        | 1.02  |
| S   | 0.0830 | 0.1039 | 2.10        | 2.64  |
| V   | 0.0177 | 0.0236 | 0.45        | 0.60  |

PIN 1. BASE  
2. EMITTER  
3. COLLECTOR

